

Project Initialization and Planning Phase

Date	26 June 2026
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	Develop an AI-powered loan approval prediction system using machine learning for accurate and quick lending decisions.
Scope	Includes data preprocessing, model training, evaluation, and deployment through a Flask web application.
Problem Statement	
Description	Traditional loan approval involves manual verification, making the process slow and prone to inconsistency.
Impact	Automation improves operational efficiency, reduces risk, and enhances customer satisfaction.
Proposed Solution	
Approach	Train Decision Tree, Random Forest, KNN, and XGBoost models using historical loan data and deploy the best model with Flask.
Key Features	• Loan eligibility prediction • Real-time prediction • User-friendly web interface • Multiple ML algorithm comparison • Fast and accurate decision support

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor specifications	Intel Core i5/i7
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, XGBoost
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Dataset	Loan Prediction Dataset (CSV)